

SECTION A - BIOLOGY

Q1. State the role of stomata in gaseous exchange and transpiration.

Q2. Why is chlorophyll essential for the process of photosynthesis?

Q3. Differentiate between aerobic and anaerobic respiration with chemical equations.

Q4. What is the function of nephrons in human kidneys?

SECTION B - CHEMISTRY & PHYSICS

Q5. Balance the chemical equation: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$

Q6. Explain why zinc reacts with dilute sulfuric acid to release hydrogen gas.

Q7. State Ohm's Law and derive the formula $V = IR$.

Q8. Calculate the equivalent resistance of two 6 ohm resistors connected in parallel.